

Data Ingestion and Analytics Pipeline Project

Description

In modern systems, a large amount of data is created from many different sources and in different formats. However, this raw data is often not clean or complete, and it is not ready for analysis. To use this data, we need to collect it, clean it, transform it, and store it in a structured way.

In this project, you will learn the basic steps of working with data. You will build a data pipeline that takes real-world data and stores it in a relational database. After that, you will use SQL to analyze the data.

This project is similar to real tasks in industry, but in a simpler form. It will help you gain practical experience and improve your skills.

Learning Objectives

By completing this project, you will learn how to:

- Work with real-world datasets
- Design a relational database schema
- Clean and transform raw data
- Build a basic ETL (Extract–Transform–Load) pipeline using Python
- Write SQL queries to analyze data
- Present insights in a clear and structured way

Dataset Selection

You must select at least one real-world dataset from an open data platform, such as:

- Kaggle
- MovieLens Dataset
- UCI Machine Learning Repository
- NYC Open Data

You may choose a dataset from any domain (e.g., e-commerce, finance, healthcare, transportation, social media, etc.), as long as it allows you to design a relational structure and perform meaningful analysis.

Tasks

1. Data Understanding

- Explore the dataset
- Identify the main entities (e.g., users, products, transactions)
- Understand the relationships between entities

2. Data Modeling

Design a relational database schema based on your dataset. Your schema must include:

- At least 3 related tables
- Primary and foreign keys
- Appropriate data types
- Constraints (e.g., NOT NULL, UNIQUE)
- Indexes where appropriate

Your design should follow basic normalization principles and avoid unnecessary data duplication.

3. Data Cleaning and Transformation

- Handle missing values
- Remove duplicate records
- Standardize formats (e.g., dates, timestamps)
- Fix incorrect data types
- Validate relationships between tables

All steps must be clearly explained in your documentation.

4. Data Pipeline (Python)

Implement a simple ETL pipeline:

- **Extract** — load raw data from files or APIs
- **Transform** — clean and prepare the data
- **Load** — insert the data into your database

Your pipeline should be organized into separate modules (e.g., `extract.py`, `transform.py`, `load.py`).

5. Database Implementation

You may use any relational database system such as:

- PostgreSQL
- MySQL
- Microsoft SQL Server

You must provide SQL scripts that create your database structure (`schema.sql`), including tables, relationships, constraints and indexes.

6. Data Analysis (SQL)

Write SQL queries to extract meaningful insights from your data.

Your analysis must include:

- Summary statistics
- Trend analysis (if applicable)
- Top or bottom entities
- Grouped analysis (e.g., by category or location)
- Multi-table queries using JOINS
- At least one real analytical question

You are expected to use: JOINS, GROUP BY, filtering, ORDER BY and at least one window function

What You Need to Submit

You must submit:

- Python pipeline code
- SQL scripts (`schema.sql`, `queries.sql`)
- Dataset description
- Explanation of data cleaning and preprocessing
- README file with:
 - Project overview
 - Dataset source
 - Pipeline description
 - Database schema explanation
 - Instructions to run the project
 - Summary of analytical results

GitHub Repository

Your project must be uploaded to GitHub and include all source code, SQL scripts and documentation.

Suggested Structure

```
project/  
  data/  
  app/
```

```
extract.py
transform.py
load.py
main.py
sql/
  schema.sql
  queries.sql
README.md
requirements.txt
```

Presentation

You will give a 10–15 minute presentation. In your presentation, you should talk about:

- Dataset and problem context
- Database design
- Pipeline workflow
- Data cleaning steps
- SQL analysis and key insights
- Challenges and solutions

Important Notes

- This project simulates real-world data tasks
- There is no single correct solution
- Your work must be original
- You should clearly explain your decisions

Work Format

You can work individually or in a group (2–4 students).

Good luck with your project!